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Cryptography Theory and Practice
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Mini Project 1

Question 1

a. The BPV algorithm works in an online/offline structure where the offline phase, with inputs N and e where N=p*q, precomputes n random values and their modular exponent stored as (x, x^e mod N). Then another security parameter T is decided, based on subset size k, each exponentiation chooses T of these pairs and multiplies the x^e mod N values. This results in every modular exponentiation only requiring T multiplications instead of 1.5L in a square-multiply method, meaning the number of multiplications needed is constant and not tied to the length of the exponent.

b. Given L=4096 and K=16 then the estimated number of multiplications needed is T=L/K=4096/16=256. This means that, given the number of multiplications for square-multiply is 1.5L, the square-multiply needs 6,144 multiplications, resulting in 5,888 multiplications saved.

Question 2

a. Since the device can store at most 127 precomputed values and for a given input K 2^K - 1 precomputed values are needed, K=7 is the largest value possible without exceeding the device's memory limit.

b. Given the approximate number of multiplications needed is T=L/K with an L=1024 and K=7, the approximate multiplications needed is 146.

c. With 146 multiplications needed for the BPV method and 1536 multiplications needed for square-multiply, 1.5*1024=1536, there is a 91.5% reduction in the number of multiplications needed.

[1] Boyko, Victor et al. “Speeding up Discrete Log and Factoring Based Schemes via Precomputations.” *International Conference on the Theory and Application of Cryptographic Techniques* (1998).